7.5 Coordinate Proofs for Quadrilaterals

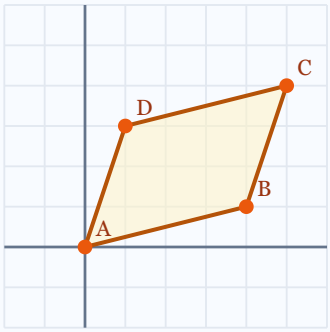
On the coordinate plane, three algebraic tools let you prove exactly what kind of quadrilateral you have — no protractor needed. **Slope** ($m = \frac{y_2 - y_1}{x_2 - x_1}$) tests parallel sides (equal slopes) and perpendicular sides (slopes multiply to -1). **Distance** ($d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$) tests congruent sides or diagonals. **Midpoint** tests whether diagonals bisect each other. Chain these together to climb the hierarchy: both pairs of opposite sides parallel ⇒ parallelogram; add perpendicular adjacent sides (or congruent diagonals) ⇒ rectangle; add congruent sides (or perpendicular diagonals) ⇒ rhombus; both ⇒ square. Only one pair of parallel sides ⇒ trapezoid. Two distinct pairs of consecutive congruent sides (with perpendicular diagonals but no parallel sides) ⇒ kite. This worksheet uses full coordinates and asks you to justify the most specific classification possible.

Examples

Example 1: Prove a parallelogram using slope

Vertices $A(0, 0)$, $B(4, 1)$, $C(5, 4)$, $D(1, 3)$. Show that $ABCD$ is a parallelogram.

Solution



Opposite sides have equal slope, so $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$.

Slope of $\overline{AB}$: $\frac{1 - 0}{4 - 0} = \frac{1}{4}$. Slope of $\overline{DC}$: $\frac{4 - 3}{5 - 1} = \frac{1}{4}$. Equal ⇒ $\overline{AB} \parallel \overline{DC}$.

Slope of $\overline{AD}$: $\frac{3 - 0}{1 - 0} = 3$. Slope of $\overline{BC}$: $\frac{4 - 1}{5 - 4} = 3$. Equal ⇒ $\overline{AD} \parallel \overline{BC}$.

Both pairs of opposite sides are parallel, so $ABCD$ is a **parallelogram**.

Example 2: Upgrade to a rectangle

In parallelogram $ABCD$, adjacent sides have slopes 2 and $-\frac{1}{2}$. Show that $ABCD$ must be a rectangle.

Solution

Two lines are perpendicular exactly when their slopes multiply to -1:

$2 \times (-\frac{1}{2}) = -1.$

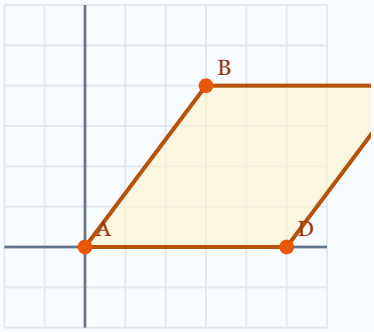
So the adjacent sides meet at a right angle. Since $ABCD$ is already a parallelogram, one right angle forces all four angles to be right angles (opposite angles congruent, consecutive angles supplementary).

Answer: $ABCD$ is a **rectangle**.

Example 3: Prove a rhombus using distance

Vertices $A(0,0)$, $B(3,4)$, $C(8,4)$, $D(5,0)$. Show that $ABCD$ is a rhombus.

Solution



All four sides have length 5.

$AB = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$

$BC = \sqrt{(8-3)^2 + (4-4)^2} = \sqrt{25} = 5.$

$CD = \sqrt{(5-8)^2 + (0-4)^2} = \sqrt{9+16} = 5.$

$DA = \sqrt{(0-5)^2 + (0-0)^2} = \sqrt{25} = 5.$

All four sides equal 5, so $ABCD$ is a **rhombus**. (It is also a parallelogram, since $\overline{AB}$ and $\overline{DC}$ are both the vector $(3,4)$.)

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Example 4: Verify a square using slope and distance together

Vertices $A(0,0)$, $B(4,0)$, $C(4,4)$, $D(0,4)$. Verify that $ABCD$ is a square.

Solution

Sides equal? $AB = 4$, $BC = 4$, $CD = 4$, $DA = 4$ — all congruent.

Right angles? Slope of $\overline{AB}$ is 0 (horizontal); slope of $\overline{BC}$ is undefined (vertical). A horizontal side and a vertical side are always perpendicular, so $\angle B = 90°$. The same reasoning applies at every vertex.

Four congruent sides and four right angles ⇒ **square**.

Example 5: Prove a trapezoid — only one pair parallel

Vertices $A(0,0)$, $B(6,0)$, $C(4,3)$, $D(1,3)$. Classify $ABCD$.

Solution

Slope of $\overline{AB}$: $\dfrac{0-0}{6-0} = 0$. Slope of $\overline{DC}$: $\dfrac{3-3}{4-1} = 0$. Equal ⇒ $\overline{AB} \parallel \overline{DC}$.

Slope of $\overline{AD}$: $\dfrac{3-0}{1-0} = 3$. Slope of $\overline{BC}$: $\dfrac{3-0}{4-6} = -\dfrac{3}{2}$. Not equal ⇒ $\overline{AD}$ and $\overline{BC}$ are *not* parallel.

Exactly one pair of opposite sides is parallel, so $ABCD$ is a **trapezoid** (with bases $\overline{AB}$ and $\overline{DC}$).

Example 6: Square from diagonals alone

A quadrilateral has diagonals from $(0,0)$ to $(6,6)$ and from $(0,6)$ to $(6,0)$. Classify the quadrilateral.

Solution

Diagonal 1 length: $\sqrt{6^2+6^2} = 6\sqrt{2}$. Diagonal 2 length: $\sqrt{6^2+6^2} = 6\sqrt{2}$ — congruent diagonals.

Slope of diagonal 1: $\dfrac{6-0}{6-0} = 1$. Slope of diagonal 2: $\dfrac{0-6}{6-0} = -1$. Product $= 1 \times (-1) = -1$ — perpendicular diagonals.

Midpoint of diagonal 1: $(3,3)$. Midpoint of diagonal 2: $(3,3)$ — same midpoint, so the diagonals bisect each other (parallelogram).

Congruent, perpendicular diagonals that bisect each other ⇒ the quadrilateral is a **square**.

Example 7: Midpoint test proves a parallelogram

$ABCD$ has diagonals $\overline{AC}$ with $A(1,2)$, $C(7,8)$, and $\overline{BD}$ with $B(2,7)$, $D(6,3)$. Show $ABCD$ is a parallelogram.

Solution

Midpoint of $\overline{AC}$: $\left(\frac{1+7}{2}, \frac{2+8}{2}\right) = (4,5)$.

Midpoint of $\overline{BD}$: $\left(\frac{2+6}{2}, \frac{7+3}{2}\right) = (4,5)$.

Both diagonals share the same midpoint $(4,5)$, so they bisect each other. A quadrilateral whose diagonals bisect each other is a **parallelogram**.

Example 8: Basic distance for adjacent sides

Points $P(1,1)$, $Q(4,1)$, $R(4,5)$, $S(1,5)$. Find PQ and PS.

Solution

$PQ = \sqrt{(4-1)^2 + (1-1)^2} = \sqrt{9} = 3$.

$PS = \sqrt{(1-1)^2 + (5-1)^2} = \sqrt{16} = 4$.

Answer: $PQ = 3$; $PS = 4$ (this is a 3-by-4 rectangle).

Example 9: Perpendicular slope check with fractions

Slopes $m_1 = \frac{2}{3}$ and $m_2 = -\frac{3}{2}$. Are the corresponding sides perpendicular?

Solution

Multiply the slopes: $\frac{2}{3} \times \left(-\frac{3}{2}\right) = -\frac{6}{6} = -1$.

Since the product is -1, the sides are **perpendicular**.

Example 10: Capstone — classify as specifically as possible

Vertices $A(0,0)$, $B(5,2)$, $C(7,7)$, $D(2,5)$. Classify $ABCD$ as specifically as possible.

Solution

Check for parallelogram (slopes): $\overline{AB}$: slope $\frac{2}{5}$; $\overline{DC}$: $\frac{7-5}{7-2} = \frac{2}{5}$ — equal. $\overline{AD}$: slope $\frac{5}{2}$; $\overline{BC}$: $\frac{7-2}{7-5} = \frac{5}{2}$ — equal. Both pairs parallel ⇒ parallelogram.

Check side lengths: $AB = \sqrt{5^2+2^2} = \sqrt{29}$; $AD = \sqrt{2^2+5^2} = \sqrt{29}$ — adjacent sides congruent, so *all four* sides are congruent (opposite sides already matched by the parallelogram property) ⇒ rhombus.

Check right angles: $\overrightarrow{AB} = (5,2)$, $\overrightarrow{AD} = (2,5)$. Dot product $= 5(2) + 2(5) = 20 \neq 0$, so the sides are *not* perpendicular — not a rectangle, so not a square.

Answer: $ABCD$ is a **rhombus that is not a square**, with side length $\sqrt{29}$.

Example 11: Coordinate proof of a kite

Vertices $A(0,3)$, $B(2,0)$, $C(0,-4)$, $D(-2,0)$. Show that $ABCD$ is a kite (and not a rhombus).

Solution

$AB = \sqrt{2^2+3^2} = \sqrt{13}$.

$AD = \sqrt{2^2+3^2} = \sqrt{13}$ — matches AB.

$BC = \sqrt{2^2+4^2} = \sqrt{20}$.

$CD = \sqrt{2^2+4^2} = \sqrt{20}$ — matches BC, but $\sqrt{20} \neq \sqrt{13}$.

Two distinct pairs of consecutive congruent sides ($AB = AD$ and $CB = CD$), with the pairs unequal to each other ⇒ $ABCD$ is a **kite**, not a rhombus (a rhombus would need *all four* sides equal).

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Parallel

Slopes $\frac{3}{4}$ and $\frac{3}{4}$. Parallel?

Answer: **Verification:**

Problem 2: Perpendicular

Slopes 4 and $-\frac{1}{4}$. Perpendicular?

Answer: **Verification:**

Problem 3: Midpoint

Diagonals share midpoint $(2, 3)$. The figure must be a...

Answer: **Verification:**

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Problem 4: Distance

$A(0, 0), B(6, 0)$. Find AB.

Answer: **Verification:**

Problem 5: Classify

A parallelogram has four congruent sides. Name it.

Answer: **Verification:**

Problem 6: Classify

A parallelogram has a right angle. Name it.

Answer: **Verification:**

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Problem 7: Trapezoid

A quadrilateral has exactly one pair of parallel sides. Name it.

Answer: **Verification:**

Problem 8: Square test

A rhombus has congruent diagonals. Name it.

Answer: **Verification:**

Problem 9: Slope

A vertical side has slope…

Answer: **Verification:**

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Problem 10: Distance

$C(2, 3), D(2, 8)$. Find CD.

Answer: **Verification:**

Problem 11: Perpendicular

Slopes $\frac{5}{2}$ and $\frac{2}{5}$. Perpendicular?

Answer: **Verification:**

Problem 12: Rectangle diagonals

In a rectangle, the diagonals are always...

Answer: **Verification:**

Problem 13: Proof idea

To prove a quadrilateral is a parallelogram using only its diagonals, you show that they…

Answer: **Verification:**

Problem 14: Challenge

Vertices $A(0,0)$, $B(4,2)$, $C(6,6)$, $D(2,4)$. Classify $ABCD$ as specifically as possible and give the side length.

Answer: **Verification:**

Problem 15: Challenge

Vertices $W(0,4)$, $X(3,0)$, $Y(0,-5)$, $Z(-3,0)$. Show $WXYZ$ is a kite and give both distinct side lengths.

Answer: **Verification:**

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✓ Answer Key

1. Yes
2. Yes
3. Parallelogram
4. 6
5. Rhombus
6. Rectangle
7. Trapezoid
8. Square
9. Undefined
10. 5
11. No (product $\neq -1$)
12. Congruent
13. Bisect each other (share the same midpoint)
14. Rhombus (not a square); side $= 2\sqrt{5}$
15. Kite; $WX = ZW = 5, XY = YZ = \sqrt{34}$

Great work — keep practising!